

MATH 10A — STRUCTURE AND CONCEPTS IN MATH I

§1.2

The Egyptian Numeration System

Chapter 1 — Numeration Systems



Today's Goal

After completing this section, you should be able to:

- Understand how an additive numeration system works.
- Read and write numbers using Egyptian numerals.

LT.2 Numeration Systems

I can convert numbers between various numeration systems including Roman, Egyptian, Mayan, Babylonian, and Hindu-Arabic.

Before There Were Digits

How would you write “one hundred” if the digits 0–9 had never been invented?

Five thousand years ago, Egyptian scribes answered that question with pictures — just seven of them.



The Seven Egyptian Glyphs

Egyptian Glyph	Description	Number
	staff	1
n	heel bone	10
⤿	spiral	100
🪷	lotus flower	1,000
👉	finger	10,000
🐸	tadpole	100,000
🧑	astonished person	1,000,000

Try It Before Any Rules

How would you write 400 with only these seven symbols?

~~~~~

Four spirals, each worth 100 — together, 400. You just used the system's only rule.



# One Rule Makes It Work

The Egyptian numeration system is an example of an *additive numeration system*.

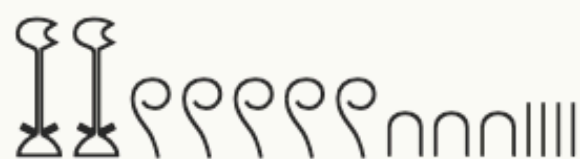
## Definition 1.2.2

An *additive numeration system* is a numeration system in which the value of a set of glyphs is the sum of the values of each glyph in the set.

The order of the glyphs helps humans read — but only the *sum* carries the meaning.

# Reading by Counting and Adding

What number is represented by the Egyptian numeral shown below?



Egyptian numerals are customarily written in decreasing order from left to right.

**Solution.** Count each glyph, multiply by its value, and add:

| Egyptian Glyph                                                                        | Count | Total Value             |
|---------------------------------------------------------------------------------------|-------|-------------------------|
|   | 2     | $2 \cdot 1,000 = 2,000$ |
|  | 5     | $5 \cdot 100 = 500$     |
|  | 3     | $3 \cdot 10 = 30$       |
|  | 4     | $4 \cdot 1 = 4$         |

Adding these gives us the Hindu-Arabic numeral 2,534.



# What We Just Did, Formalized

## **Procedure 1.2.3. Converting an Egyptian Numeral to the Corresponding Hindu-Arabic Numeral**

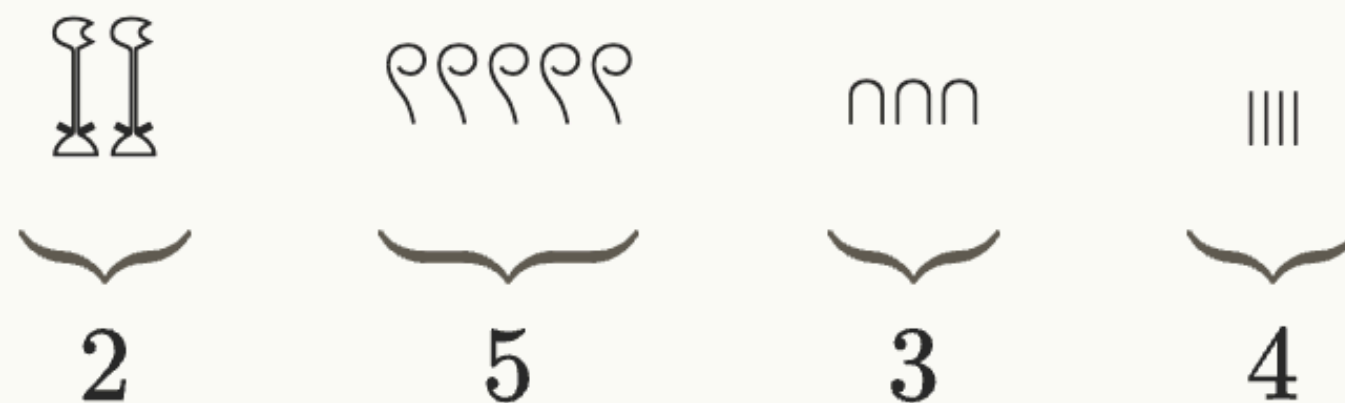
To convert an Egyptian numeral to the corresponding Hindu-Arabic numeral, do the following.

1. Identify which Egyptian glyphs (see Table 1.2.1) are being used.
2. Count the number of each different Egyptian glyph that is being used.
3. For each different Egyptian glyph used, multiply the value of the Egyptian glyph by how many of them were used.
4. Add all of the subtotals from Step 3.



# A Tempting Shortcut

A shortcut suggests itself: *“The glyphs are already listed in decreasing order — so just count how many of each there are:”*



*“Therefore, the number is 2,534.”*

**It worked this time. What could go wrong?**

# Putting the Shortcut to the Test

What number is represented by this Egyptian numeral? Try the shortcut, then Procedure 1.2.3.



**Solution.** The shortcut first: count how many of each different glyph appear, reading left to right.

 fingers: 4

 spirals: 2

 heel bones: 6

 staffs: 3

Put those counts side by side and the shortcut reads the numeral as 4,263.

# Checking with Procedure 1.2.3

The shortcut read this numeral as 4,263. Time to check its work.



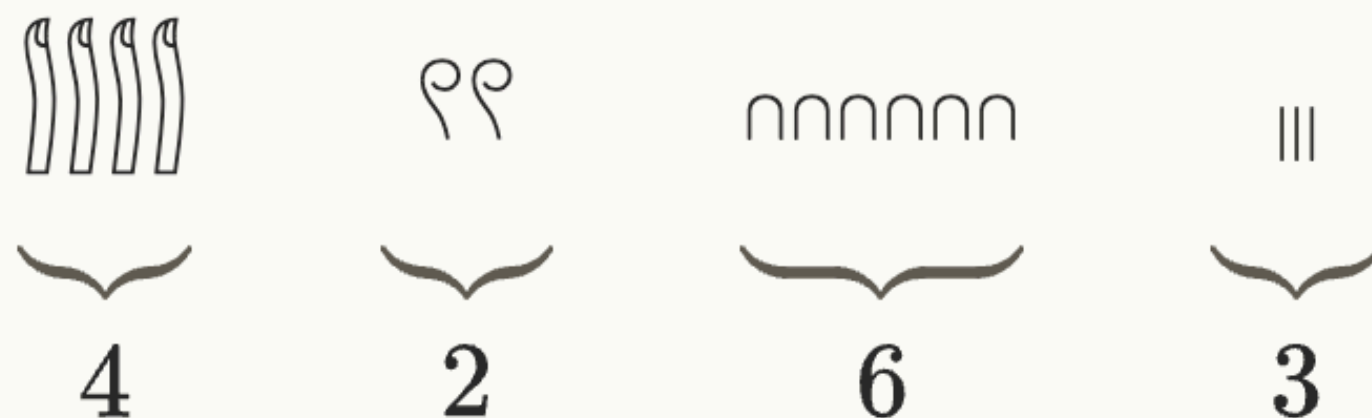
**Solution, continued.** Count each glyph, multiply by its value, and add:

| Egyptian Glyph                                                                        | Count | Total Value               |
|---------------------------------------------------------------------------------------|-------|---------------------------|
|   | 4     | $4 \cdot 10,000 = 40,000$ |
|  | 2     | $2 \cdot 100 = 200$       |
|  | 6     | $6 \cdot 10 = 60$         |
|  | 3     | $3 \cdot 1 = 3$           |

Adding gives 40,263 — but the shortcut said 4,263.

# Why the Shortcut Fails

The shortcut counts each different glyph and strings the counts together:



Read as digits, the counts say “4,263” — but the numeral is 40,263.

The counts 4, 2, 6, 3 say nothing about each glyph’s *value* — four fingers are worth 40,000, and no glyph marks the empty thousands place. Ignoring the value of each glyph that is used will cause problems.

# Writing Goes Through Expanded Form

## **Procedure 1.2.8. Converting a Hindu-Arabic Numeral to the Corresponding Egyptian Numeral**

To convert a Hindu-Arabic numeral to the corresponding Egyptian numeral, do the following.

1. Write the number in expanded form.
2. Use the expanded form to determine how many of each Egyptian glyph is needed.



# Expanded Form Does the Work

Write the number 24,153 as an Egyptian numeral.

**Solution.** We begin by writing 24,153 in expanded form.

$$24,153 = 20,000 + 4,000 + 100 + 50 + 3.$$

| Part                      | Egyptian Glyph                                                                        |
|---------------------------|---------------------------------------------------------------------------------------|
| $20,000 = 2 \cdot 10,000$ |    |
| $4,000 = 4 \cdot 1,000$   |    |
| $100 = 1 \cdot 100$       |   |
| $50 = 5 \cdot 10$         |  |
| $3 = 3 \cdot 1$           |  |



# Assembling the Numeral

Each part of the expanded form became one glyph group. Combining these gives us the Egyptian numeral



Reading the numeral back, each glyph group matches one part of the expanded form:  $20,000 + 4,000 + 1,000 + 300 + 50 + 3 = 24,153$ .

# Now You Try: Reading

## You Try 1.2.1.

What number does this Egyptian numeral represent?



**Answer.**  $100,000 + 3,000 + 20 + 4 = 103,024$ . The glyphs that do not appear contribute 0.

# Now You Try: Writing

## You Try 1.2.2.

Write 20,507 as an Egyptian numeral.

**Answer.** Two fingers, five spirals, seven staffs — no lotus flowers or heel bones needed:

